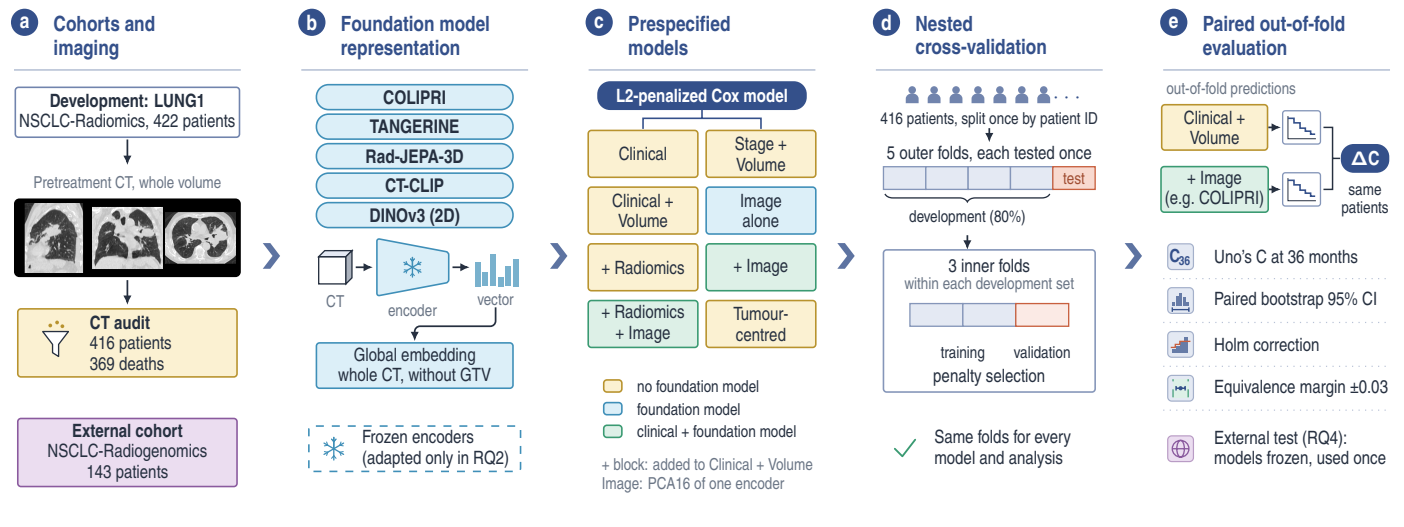

a
Cohorts and imaging
Development: LUNG1
NSCLC-Radiomics, 422 patients
Pretreatment CT, whole volume
CT audit
416 patients
369 deaths
External cohort
NSCLC-Radiogenomics
143 patients
b
Foundation model representation
COLIPRI
TANGERINE
Rad-JEPA-3D
CT-CLIP
DINOv3 (2D)
CT
encoder
vector
Global embedding
whole CT, without GTV
Frozen encoders
(adapted only in RQ2)
c
Prespecified models
L2-penalized Cox model
Clinical
Stage + Volume
Clinical + Volume
Image alone
+ Radiomics
+ Image
+ Radiomics + Image
Tumour-centred
no foundation model
foundation model
clinical + foundation model
+ block: added to Clinical + Volume
Image: PCA16 of one encoder
d
Nested cross-validation
416 patients, split once by patient ID
5 outer folds, each tested once
test
development (80%)
3 inner folds
within each development set
training
validation
penalty selection
Same folds for every model and analysis
e
Paired out-of-fold evaluation
out-of-fold predictions
Clinical + Volume
+ Image (e.g. COLIPRI)
ΔC
same patients
Uno's C at 36 months
Paired bootstrap 95% CI
Holm correction
Equivalence margin ±0.03
External test (RQ4): models frozen, used once